

Orders, Groups, and Fields

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Throughout this article p denotes a prime, and q a prime power.

1 Introduction

One of the first landmark theorems you learn in number theory is *Fermat's Little Theorem*, which states that $p \mid a^p - a$ for all primes p and integers a . The key to this theorem is noting that multiplication by a induces a *permutation* on the residues modulo p . Notably, in the case where $p \nmid a$, we find that $a^{p-1} \equiv 1 \pmod{p}$, which we may use to conclude that the *order* of a modulo p divides $p - 1$. Fermat's little theorem and the idea of orders thereafter become tremendously useful concepts for attacking number theory problems.

It turns out that both these ideas fit into a much larger framework of *group theory*. This is the study of abstract sets with a multiplicative structure, which vastly generalize some notions of multiplication you're used to (e.g. nonzero rationals, invertible matrices, permutations!) In number theory, there are two (really three) abstract algebraic notions that you should really care about:

1. the group, which describes a set with a certain multiplicative structure, and;
2. the ring / field, which describe a set with multiplicative and additive structures.

Today for simplicity we will deal with two specific cases: the *abelian* group and the field. We'll discuss what these two terms mean soon, and how they relate to number theory.

2 Groups and Orders

Definition 2.1. A *group* is a set G equipped with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following properties:

1. there is an element $e \in G$ (the identity, also denoted 1_G) such that $e \cdot g = g \cdot e = g$ for all $g \in G$;
2. each $g \in G$ admits an element $h \in G$ (denoted $h = g^{-1}$, its *inverse*) such that $g \cdot h = h \cdot g = e$;
3. the operation is associative, i.e. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$. Thus we will generally abbreviate $a \cdot b = ab$, and the above shows abc is well-defined.

Remember that when specifying a group, you must also specify its group operation! Some examples of groups:

1. $\mathbb{Z}$ equipped with addition.
2. $\mathbb{Z}/n\mathbb{Z}$ equipped with addition.
3. $\mathbb{R}$ equipped with addition.
4. $\mathbb{R} \setminus \{0\}$ equipped with multiplication.
5. $(\mathbb{Z}/p\mathbb{Z}) \setminus \{0\}$ equipped with multiplication (this will typically be denoted $(\mathbb{Z}/p\mathbb{Z})^\times$).
6. The set of invertible $n \times n$ real matrices $\text{GL}_n(\mathbb{R})$ equipped with matrix multiplication.

Example 2.2. Pick an example above, and check that it indeed satisfies the group axioms.

Example 2.3. The following sets equipped with operations are all *not* groups. Can you say why?

1. $\mathbb{R}$ equipped with multiplication.
2. $\mathbb{Z}^+$ equipped with multiplication.
3. $(\mathbb{Z}/m\mathbb{Z}) \setminus \{0\}$ equipped with multiplication, where m is not a prime.
4. The set of $n \times n$ real matrices $\text{Mat}_n(\mathbb{R})$ equipped with matrix multiplication.

The group $\text{GL}_n(\mathbb{R})$ is special among the above groups in that it is not *commutative*; in general, $n \times n$ matrices A, B do not satisfy $AB = BA$. We call a group *abelian* if its group operation satisfies $ab = ba$ for all $a, b \in G$ (that is, a and b always *commute*). From now on, we will exclusively work with abelian groups today.

2.1 Group Maps

How do we compare groups, and how do we say when two groups are the same?

Definition 2.4. Given two groups G, H , a *group homomorphism* is a map of sets $\phi : G \rightarrow H$ which satisfies that for any $g, h \in G$,

$$\phi(g) \cdot \phi(h) = \phi(g \cdot h).$$

Here, on the left, the product is taken in H , while on the right, the product (inside the bracket) is taken in G .

Example 2.5. Prove that a group homomorphism, by definition, must map 1_G to 1_H .

Example 2.6. Consider two relatively prime integers m, n . Prove that any homomorphism $\phi : \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ must map every element to the identity.

Definition 2.7. A group homomorphism $\phi : G \rightarrow H$ is called an *isomorphism* if it is bijective.

In the future we will use $G \cong H$ to denote that G and H are isomorphic.

Example 2.8. Prove that given a group isomorphism ϕ , the inverse map ϕ^{-1} is also a group isomorphism. (This is how we should really define group isomorphisms; by saying there is a group homomorphism ψ such that ϕ and ψ are inverses).

2.2 Orders

Definition 2.9. Consider a group G . The *order* of an element $a \in G$ is the smallest integer $m > 0$ such that $a^m = 1_G$. If no such integer exists, then say that a has *infinite order*.

Example 2.10. How does this definition of order relate to the notion of “order modulo p ”?

Example 2.11. From your number theory experience, reconstruct the following proofs:

1. In a finite group G , every element must have finite order.
2. Given an element a with finite order m , the integers n for which $a^n = 1_G$ are precisely the multiples of m .
3. In a finite *abelian* group G , the order of any element must divide the cardinality of G . (In fact this is true for non-abelian groups, by looking at subgroups and cosets. But we won't need this.)

I trust that you already know why orders are so important. Some practice with orders is given later.

2.3 Group Constructions

Here are two ways to construct more groups:

Definition 2.12. Given a group $(G, \cdot)$, a subgroup is a subset $H \subset G$ such that $1_G \in H$, and H is closed under taking inversions and $\cdot$ (i.e. for $g, h \in H$, $gh \in H$ and $g^{-1} \in H$).

Observe that for any such subset H , $(H, \cdot)$ then forms a group on its own.

Example 2.13. Compute all the subgroups of $(\mathbb{Z}, +)$.

Definition 2.14. Given two groups G, H , define the group $G \times H$ to have underlying set given by pairs $\{(g, h) : g \in G, h \in H\}$, and to have group operation

$$(g, h) \cdot (g', h') = (g \cdot g', h \cdot h').$$

Example 2.15. Prove that $\mathbb{Z}/2 \times \mathbb{Z}/3 \cong \mathbb{Z}/6$, but $\mathbb{Z}/2 \times \mathbb{Z}/2 \not\cong \mathbb{Z}/4$.

The cartesian product $\times$ gives a useful way for constructing more groups. In fact, we may use it to construct all finite abelian groups!

Theorem 2.16 (Classification of Finite Abelian Groups). *Every finite abelian group is isomorphic to a cartesian product of the form*

$$\mathbb{Z}/n_1\mathbb{Z} \times \dots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1, \dots, n_k$ are (not necessarily distinct) positive integers.

The proof is omitted. You can construct an ad-hoc proof yourself, but maybe this theorem becomes more intuitive when you learn some ring theory.

Remark 2.17. This theorem should really be stated for *finitely generated abelian groups*, but we won't need that generality today.

Remark 2.18. It turns out that the classification of finite groups is *much, much harder*. At the beginning of this millenium, the classification of finite *simple* groups was finally completed, consisting of hundreds of papers, each dozens of pages long. Even now, humanity's knowledge of this fact is considered dubious, since the proof is so gargantuan.

2.4 Exercises on Groups

2.4.1 Non-abelian groups

1. Given any elements $x_1, \dots, x_n$ of a group G , prove that

$$(x_1 x_2 \dots x_n)^{-1} = x_n^{-1} x_{n-1}^{-1} \dots x_1^{-1}.$$

2. Show any finite group G of even order contains an element $x \in G$ such that $x \neq 1_G$ but $x^2 = 1_G$.
3. For any $a \in G$, let $\phi_a : G \rightarrow G$ be the map taking $x \rightarrow axa^{-1}$.
 - (a) Let an *automorphism* of G be an isomorphism from G to itself. Prove that the automorphisms $\text{Aut}(G)$ form a group under composition, and prove $\phi_a \in \text{Aut}(G)$.
 - (b) Show that the map $\phi : G \rightarrow \text{Aut}(G)$ given by $a \mapsto \phi_a$ is a homomorphism.
 - (c) Prove that given any isomorphism φ , $\varphi(x)$ has the same order as x . Hence this is also true for ϕ_a .
4. ★ Consider four elements a, b, c, d of a finite (not necessarily abelian) group G , satisfying

$$bab^{-1} = a^2;$$

$$cbc^{-1} = b^2;$$

$$dcd^{-1} = c^2;$$

$$ada^{-1} = d^2.$$

Prove that a, b, c, d all equal the identity.

5. Consider any two elements a, t of a finite group G satisfying

$$tat^{-1}ata^{-1}t^{-1}a^{-2} = 1_G.$$

Prove that $a = 1_G$.

6. ★ An important class of groups we did not touch above are the *permutation groups*. Explicitly, let the group S_n be the set of permutations on n elements, where the group operation is composition.

- Prove that this is indeed a group.
- Given any subset $S = \{s_1, \dots, s_n\} \subset G$, we say that S *generates* G if every element of G is expressible as a product of the elements $\{s_1, s_1^{-1}, \dots, s_n, s_n^{-1}\}$.
For distinct integers $i, j \in [1, n]$, let (ij) denote the transposition swapping the i^{th} and j^{th} elements. Show that the transpositions $(12), (23), \dots, (n-1\ n)$ generate S_n . Can you find a smaller generating set of S_n ?
- Prove that every finite *abelian* group is a subgroup of S_n for some integer n .
- Prove that every finite group is a subgroup of S_n for some integer n . (As a hint, consider the permutations arising by a group “acting on itself”). This is known as Cayley’s theorem.

2.4.2 Abelian Groups

1. Given any two positive integers a, b , prove

$$\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z} \cong \mathbb{Z}/\gcd(a, b)\mathbb{Z} \times \mathbb{Z}/\text{lcm}(a, b)\mathbb{Z}.$$

2. Prove that for any finite abelian group G , there exists a unique tuple $(a_1, \dots, a_n)$ of positive integers such that $a_1 \mid a_2 \mid \dots \mid a_n$, and

$$G \cong \mathbb{Z}/a_1\mathbb{Z} \times \dots \times \mathbb{Z}/a_n\mathbb{Z}.$$

3. ★ Consider an *abelian* group G and a subgroup $H \subset G$. For any $a \in G$, let aH denote the subset $\{ah : h \in H\} \subset G$ (we will call aH a *coset* of H).

- Show that one can give the cosets aH a group structure, so that they form a group, which we denote G/H .
- What goes wrong when G is not abelian? What condition do we need to form the quotient G/H in this case?

2.4.3 Contest Problems

- Is there a finite abelian group G such that the product of the orders of all its elements is 2^{2009} ?
- Suppose that G is a finite group generated by the two elements g and h , where the order of g is odd. Show that every element of G can be written in the form

$$g^{m_1}h^{n_1}g^{m_2}h^{n_2}\dots g^{m_r}h^{n_r}$$

with $1 \leq r \leq |G|$ and $m_n, n_1, m_2, n_2, \dots, m_r, n_r \in \{1, -1\}$. (Here $|G|$ is the number of elements of G .)

- Determine which positive integers n have the following property: For all integers m that are relatively prime to n , there exists a permutation $\pi : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ such that $\pi(\pi(k)) \equiv mk \pmod{n}$ for all $k \in \{1, 2, \dots, n\}$.
- Let p be an odd prime number such that $p \equiv 2 \pmod{3}$. Define a permutation π of the residue classes modulo p by $\pi(x) \equiv x^3 \pmod{p}$. Show that π is an even permutation if and only if $p \equiv 3 \pmod{4}$.

If you have time left over, here is a fun problem testing your skill with orders.

- Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients, such that no nonconstant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)} - 1$ divides $3^{P(n)} - 1$. Prove that $Q(x)$ is a constant polynomial.

3 Fields

Groups are great partially because they're very simple (as far as abstract algebraic objects go). But this also means that they're by themselves inadequate for encompassing most of the arithmetic operations we deal with day-to-day. Notably, the most important arithmetic object in mathematics is $\mathbb{Z}$, and this has *two* interacting operations. To capture this structure, we will introduce the notions of a *ring* and *field*.

3.1 Starting Examples

Definition 3.1. A *field* K is a set equipped with *two* operations $+, \times$ such that:

1. $(K, +)$ is an abelian group with identity 0;
2. $(K \setminus \{0\}, \times)$ is an abelian group with identity 1, and;
3. multiplication and addition are distributive: $a \times (b + c) = a \times b + a \times c$.

A corollary of 3) is that $a \times 0 = 0$, and if $a \times b = 0$ then $0 \in \{a, b\}$. This is required for 2) to be well-defined.

Example 3.2. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, and $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ are fields, while $\mathbb{Z}, \mathbb{Z}[x], \mathbb{Q}[x], \mathbb{Z}/p^2\mathbb{Z}$, and $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ are not. Here the cartesian product is defined analogously to the group case.

Definition 3.3. *Rings* are defined from fields by dropping some conditions on the multiplication $\times$:

1. If we do not require elements to have multiplicative inverses in $(K \setminus \{0\}, \times)$ (but keep the commutativity), then K becomes a *commutative ring*. $\mathbb{Z}, \mathbb{Z}[x]$, and $\mathbb{Q}[x]$ are examples of commutative rings.
2. If we further do not require $\times$ to be commutative, then K becomes a (*non-commutative*) *ring*. Here we now must require the distributivity law for right products: $(b + c) \times a = b \times a + c \times a$. $\text{Mat}_{n \times n}(\mathbb{R})$ is an example of a non-commutative ring.
3. If we further do not require $\times$ to admit a multiplicative identity 1, then K becomes a *rng* (terminology is because the “i” is dropped). $2\mathbb{Z}$ is an example of a rng.

Today we will mostly focus on fields, though examples of commutative rings will occasionally pop up. You won't need to think about non-commutative rings.

As in the group case, in order to define when two fields are the same, we need to define the notion of a field homomorphism:

Definition 3.4. Given two fields K, L , a *field homomorphism* between them is a map $\phi : K \rightarrow L$ such that for all $a, b \in K$,

$$\begin{aligned}\phi(1_K) &= 1_L; \\ \phi(a + b) &= \phi(a) + \phi(b); \\ \phi(ab) &= \phi(a)\phi(b).\end{aligned}$$

Definition 3.5. A field *isomorphism* is a bijective field homomorphism.

Example 3.6. Some rudimentary checks:

1. Prove that the inverse of a field isomorphism is also a field isomorphism.
2. Prove that every field homomorphism is injective.

3.2 Field Extensions

Unlike in the group case, the cartesian product of two fields is not a field. So we'll need a new recipe for constructing interesting fields.

One famous example of such a construction is the complex numbers $\mathbb{C}$, which we constructed from $\mathbb{R}$ by adjoining an element i satisfying $i^2 = -1$. This equation $x^2 + 1 = 0$ had no solutions before, so $i \notin \mathbb{R}$, but the identity $i^2 = -1$ allowed us to find that $\mathbb{R} \oplus i\mathbb{R}$ is closed under multiplication, and in fact formed a field of its own.

To emulate this process for other fields, we'll need some theory. First, a formal definition:

Definition 3.7. Given any field K , let $K[x]$ be the set of one-variable polynomials (in the indeterminate x) with coefficients in K . Explicitly, these are formal linear combinations of the form

$$a_0 + a_1x + \dots + a_nx^n$$

where $a_0, \dots, a_n \in K$ (and n is some nonnegative integer).

Remark 3.8. By evaluating x to be any element of K , we can interpret any element of $K[x]$ as a function on K . However, **be warned that polynomials are first and foremost formal objects!** The reason we do this is because for finite fields, there will exist nonzero polynomials (e.g. $x^p - x$ for $\mathbb{F}_p$) which are identically zero as functions, so we want to distinguish these polynomials from the zero polynomial.

As you may recognize from the case of rational or real polynomials, $K[x]$ satisfies many useful properties. For instance, I can perform the *Euclidean algorithm* on any pair of polynomials (p, q) ; that is, I can produce polynomials a, r such that $p(x) = q(x)a(x) + r(x)$, where the degree of r is less than the degree of q . Notably, this allows me to define the gcd of any two polynomials, and we will know (by Bezout) that $\gcd(p, q)$ is expressible as $bp + cq$ for some polynomials b, c . This further lets me factor a polynomial uniquely into a collection of *irreducible* polynomials:

Definition 3.9. An *irreducible polynomial* in $K[x]$ is a monic polynomial of nonzero degree which is not divisible by any other monic polynomial of nonzero (lesser) degree.

Using this, we may generalize our construction of $\mathbb{C}$ as follows.

Example 3.10. Given any field K and polynomial $p(x) \in K[x]$, define $K[x]/(p(x))$ to be the polynomials in $K[x]$ modulo $p(x)$ (where $a(x) \equiv b(x) \pmod{p(x)}$ if $p(x) \mid a(x) - b(x)$). Then $K[x]/(p(x))$ is a commutative ring, and if $p(x)$ is an *irreducible* polynomial then $K[x]/(p(x))$ is a field. Indeed, $K[x]/(p(x))$ inherits its $+$ and $\times$ properties from $K[x]$, so it suffices to check that nonzero elements in $K[x]/(p(x))$ have multiplicative inverses. But this is true since if $p \nmid q$, then by Bezout there exist a, b such that $ap + bq = \gcd(p, q) = 1$, so $bq \equiv 1 \pmod{p}$.

Example 3.11. Let $K = \mathbb{R}$. The polynomial $x^2 + 1$ is irreducible in $\mathbb{R}[x]$, so $\mathbb{R}[x]/(x^2 + 1)$ forms a field. In particular, this field is isomorphic to $\mathbb{C}$, explicitly via the isomorphism sending x to i . Similarly, $x^2 + 1$ is irreducible in $\mathbb{Q}[x]$, and $\mathbb{Q}[x]/(x^2 + 1)$ is the field of *Gaussian rationals*.

Remark 3.12. Given a field K , and an element α which is abstractly a root of some irreducible polynomial $p(x) \in K[x]$, we will denote $K[\alpha] : K[x]/p(x)$.

Example 3.13. The polynomial $x^2 - 2$ is irreducible in $\mathbb{Q}[x]$, so the quotient $\mathbb{Q}[x]/(x^2 - 2) = \mathbb{Q}[\sqrt{2}]$ is a field. On the level of vector spaces, it's equal to $\mathbb{Q} \oplus \mathbb{Q}\sqrt{2}$.

3.3 Finite Fields

Aside from $\mathbb{Q}$, one of the most important fields in number theory are the *finite fields*. You already know a collection of finite fields: $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, the field of integers mod p (we'll use $\mathbb{F}_p$ to emphasize the field structure). However, it turns out that these are not the only finite fields!

Example 3.14. Let $K = \mathbb{F}_2$. The polynomial $x^2 + x + 1$ is irreducible in $\mathbb{F}_2[x]$, so $\mathbb{F}_2[x]/(x^2 + x + 1)$ forms a field. Notably, by the Euclidean algorithm, each polynomial in $\mathbb{F}_2[x]$ has a unique representative in $\mathbb{F}_2[x]/(x^2 + x + 1)$ of degree ≤ 1 . There are four such polynomials in $\mathbb{F}_2[x]$, so thus $\mathbb{F}_2[x]/(x^2 + x + 1)$ is a field with four elements (which we denote $\mathbb{F}_4$).

Example 3.15. Prove that given a degree k irreducible polynomial $r(x)$ in $\mathbb{F}_p[x]$, we have that $\mathbb{F}_p[x]/(r(x))$ is a field with p^k elements.

So the theory of finite fields may be richer than we thought! Here are two questions we need to answer in order to classify the finite fields:

1. For which q does there exist a finite field with q elements?
2. For each q , how many finite fields are there with q elements (up to isomorphism)?

It turns out that answering 1) is not that hard:

Definition 3.16. The *characteristic* of a field is the smallest positive integer n such that

$$\underbrace{1 + \dots + 1}_{n \text{ ones}} = 0.$$

If no such n exists, we say the field has characteristic 0.

Example 3.17. Prove the characteristic of a field, if nonzero, must be prime.

In fact, given a field with characteristic p , we have that the set of integer multiples of 1 form a subfield isomorphic to $\mathbb{F}_p$. Every field K of $q < \infty$ elements must have nonzero characteristic by Pigeonhole, so it must have a “canonical” copy of $\mathbb{F}_p$ sitting inside it.

In fact, for those of you who know a bit of linear algebra, this already gives a significant restriction on the properties of a finite field.

Theorem 3.18. *A finite field must have cardinality a prime power.*

Proof. Suppose it has characteristic p . Consider only the multiplication structure it inherits from multiplication by integers. Viewed this way, it forms a vector space over $\mathbb{F}_p$, and thus has cardinality a power of p . (In general, if K is a subfield of L , then multiplication by K makes L into a K -vector space. So if K, L are both finite, then $|L|$ is a power of $|K|$). $\square$

As a warning, note that $\mathbb{F}_q$ has characteristic p , not q (and is thus not isomorphic to $\mathbb{Z}/q\mathbb{Z}$ as an additive abelian group!) Anyways this answers our question 1). We now show that in fact, each prime power q admits *exactly one* finite field of cardinality q .

3.4 $\mathbb{F}_q$ and its properties

This section is based on *Artin's Algebra*, Section 15.7. Throughout this section, set $q = p^k$.

Constructing *one* example of a finite field with q elements theoretically should not be that hard. After all, the above example indicates that it's equivalent to finding an irreducible polynomial in $\mathbb{F}_p[x]$ of degree k . However it's not entirely clear why such an irreducible polynomial must exist, and it's *definitely* not clear why any two choices of irreducible polynomial lead to the same resulting $\mathbb{F}_q$. Thus we will instead approach constructing $\mathbb{F}_q$ differently, and for this it will be helpful to first understand what kinds of properties $\mathbb{F}_q$ *should* have.

First, note that every element of $\mathbb{F}_q$ satisfies some polynomial relation over $\mathbb{F}_p$, i.e. it is the root of some polynomial in $\mathbb{F}_p[x]$.

Example 3.19. Prove this using pigeonhole. As an optional exercise, bound the degree of this polynomial.

Now, an important property about $\mathbb{F}_p$ is that its multiplicative structure is very nice. Indeed, for any field K , let $K^\times$ denote $K \setminus \{0\}$ viewed as a multiplicative group. Then the existence of primitive roots is equivalent to $\mathbb{F}_p^\times$ being cyclic. In fact we can generalize this as follows:

Theorem 3.20. *Consider any finite field $\mathbb{F}_q$ of q elements. Then $\mathbb{F}_q^\times$ is cyclic.*

This proof is very cool :)

Proof. The group $\mathbb{F}_q^\times$ is abelian, so by our classification of finite abelian groups, we may express it as

$$\mathbb{F}_q^\times \cong \mathbb{Z}/a_1\mathbb{Z} \times \dots \times \mathbb{Z}/a_n\mathbb{Z}$$

for some positive integers $a_1 \mid \dots \mid a_n$. This implies that the cardinality of $\mathbb{F}_q^\times$ is $q - 1 = a_1 a_2 \dots a_n$, and every element of $\mathbb{F}_q^\times$ has order dividing a_n .

Now, the latter remark implies that the polynomial $x^{a_n} - 1$ has $q - 1$ roots in $\mathbb{F}_q$. But by factoring out these roots in $\mathbb{F}_q[x]$, this implies that $x^{a_n} - 1$ has degree at least $q - 1$. Thus $a_n \mid q - 1 \leq a_n$, so $a_n = q - 1$. Hence $\mathbb{F}_q^\times \cong \mathbb{Z}/(q - 1)\mathbb{Z}$, as desired. $\square$

As a sanity check, note that $p - 1 \mid q - 1$, so the order of any element in $\mathbb{F}_p^\times$ divides $q - 1$ as well. We should expect this since $\mathbb{F}_p$ sits inside $\mathbb{F}_q$.

Before we prove our main theorem, let's do some practice.

Example 3.21. Pick one of the two examples to do:

1. Find an irreducible degree 3 polynomial $p(x)$ in $\mathbb{F}_2[x]$. Let α be one of its roots; then express every element of $\mathbb{F}_2[x]/(p(x))$ in terms of α , and identify one element which generates $(\mathbb{F}_2[x]/(p(x)))^\times$.
2. Find an irreducible degree 2 polynomial $p(x)$ in $\mathbb{F}_3[x]$. Let β be one of its roots; then express every element of $\mathbb{F}_3[x]/(p(x))$ in terms of β , and identify one element which generates $(\mathbb{F}_3[x]/(p(x)))^\times$.

With this, we may now prove the following theorem:

Theorem 3.22. *For every prime power $q = p^k$, $\mathbb{F}_q$ exists and is unique.*

This proof is pretty technical, so if you don't understand everything on a first pass that's totally okay.

Proof. We first prove existence. First we produce a (finite) field extension over $\mathbb{F}_p$ in which $x^q - x$ splits into linear factors, as follows: repeatedly take a remaining irreducible factor $r(x)$ of $x^q - x$ in the current field L , replace L with $L[x]/(r(x))$, and repeat. Call the field we produce K . We claim that the subset S of roots of $x^q - x$ in K form a field of cardinality q . We need to check:

1. S contains 1, which is clear;
2. $a \in S$ implies $a^{-1} \in S$, which is clear;
3. if $a, b \in S$, then so is ab ; also clear;
4. if $a, b \in S$ then so is $a + b$; this is because $(a + b)^p = a^p + b^p$ (in $\mathbb{F}_q$) by the Binomial theorem, hence $(a + b)^q = a^q + b^q$ by induction;
5. S actually has q elements. To do this, it suffices to show that in any field K of characteristic p , the roots of $x^q - x$ are distinct. But this is because the (formal) derivative of $x^q - x$ in $\mathbb{F}_p[x]$ is -1 , so $x^q - x$ shares no roots with its formal derivative. Hence the roots of $x^q - x$ are distinct.

So $\mathbb{F}_q$ exists. Now given any field K with q elements, pick any generator α of the cyclic group $\mathbb{F}_q^\times$. Then α must be the root of a polynomial in $\mathbb{F}_p[x]$, as before; consider its minimal polynomial $f(x)$. We have that all powers of α are in $\mathbb{F}_p[x]/(f(x))$, so $\mathbb{F}_p[x]/(f(x)) \cong K$. Moreover, since α vanishes under $x^q - x$, we find that $f(x) \mid x^q - x$ as well. Thus we find that every field of q elements is of the form $\mathbb{F}_p[x]/(f(x))$ for some irreducible degree k polynomial $f(x) \in \mathbb{F}_p[x]$ (which must divide $x^q - x$), and $x^q - x$ always splits into a product of linear factors in $\mathbb{F}_p[x]/(f(x))$. Given any other irreducible degree k factor $g(x)$ of $x^q - x$, since $g(x)$ also splits completely in $\mathbb{F}_p[x]/(f(x))$, we may pick any root β of it in $\mathbb{F}_p[x]/(f(x))$, and note that $\mathbb{F}_p[x]/(g(x)) \cong \mathbb{F}_p[\beta] = \mathbb{F}_p[x]/(f(x))$. This finishes. $\square$

Example 3.23. In your previous example, either factor $x^8 - x$ completely in $\mathbb{F}_2$ or $x^9 - x$ completely in $\mathbb{F}_3$.

3.5 Field Theory, Intuitively

We've now discussed a good amount of field theory. What we've observed (especially in the finite field case) is that fields share a lot of properties, especially field extensions of a given field (such as $\mathbb{F}_p$ and $\mathbb{F}_q$). Notably, though $x^2 + 1$ is irreducible in $\mathbb{Q}$ or $\mathbb{F}_3$, it contains roots in $\mathbb{Q}[i]$ or $\mathbb{F}_9$, and we shouldn't treat these fields as totally different from their predecessors. The lesson you should take away is thus that such irreducible expressions, when they appear in number theory, are *not so alien!* You can still work with them and factor them, and this may only mean that you have to pass up to a bigger field.

Here's a contest example of this ethos:

Example 3.24. [ISL 2003 N7] The sequence $a_0, a_1, a_2, \dots$ is defined as follows:

$$a_0 = 2, \quad a_{k+1} = 2a_k^2 - 1 \quad \text{for } k \geq 0.$$

Prove that if an odd prime p divides a_n , then 2^{n+3} divides $p^2 - 1$.

Proof. First, let's get a grasp on the sequence

$$x \mapsto 2x^2 - 1.$$

This relation maps $\cos \theta \mapsto \cos(2\theta)$, so if we pick an x_0 such that $x_0 + 1/x_0 = 4$, and define $x_{i+1} = x_i^2$; then we will always have $x_i + 1/x_i = 2a_i$ by induction.

The issue is that x_i is not guaranteed to exist in $\mathbb{F}_p$. Indeed, we want $x_0 = 2 + \sqrt{3}$, which only exists in $\mathbb{F}_p$ if 3 is a quadratic residue. But *not to worry*; there always exists an $x_i \in \mathbb{F}_{p^2}$ which satisfies this. So we can work exclusively in $\mathbb{F}_{p^2}$ and solve the problem there.

Now, we have that by assumption that $x_n + 1/x_n = 0$ in $\mathbb{F}_p$, so $x_n^2 + 1 = 0$. Hence x_n has order 4, and so x_0 has order 2^{n+2} . This almost solves the problem, as it implies $2^{n+2} \mid p^2 - 1$, yet we want $2^{n+3} \mid p^2 - 1$. But not to worry; it turns out that $x_0 = 2 + \sqrt{3}$ is a square in $\mathbb{F}_{p^2}$; indeed

$$2 + \sqrt{3} = \left(\frac{1 + \sqrt{3}}{\sqrt{2}} \right)^2,$$

and both $1 + \sqrt{3}$ and $\sqrt{2}$ exist in $\mathbb{F}_{p^2}$. So $x = \frac{1 + \sqrt{3}}{\sqrt{2}}$ has order 2^{n+3} in $\mathbb{F}_{p^2}$, and so $2^{n+3} \mid p^2 - 1$ as desired. $\square$

3.6 Final Thoughts

Today's lecture was much more systematic than previous lectures, and maybe gave you some insight into what higher math is all about. To me, field theory is a great showcase of two very important aspects of higher math that are not as prevalent in contest math:

1. a willingness to work with the *impossible*, and
2. a desire to study things systematically.

Contest problems are by and large about applying your existing tools in creative ways to study problems just barely out of your reach. But sometimes to understand what's actually going on, we have to dig much deeper, and also be more brave to work with concepts that may previously seem alien. This is most prevalent in number theory, where many problems are about polynomials and exponents: olympiad math gives you some tricks to deal with these problems, but to really understand how these expressions work, you have to brave the land of much more unfamiliar territory (e.g. ring and field theory). Hopefully the examples today gave you a glimpse of how powerful these methods can be. If you're brave enough, maybe you'd like to try your hand at pioneering research in the future!

3.7 Exercises on Fields

3.7.1 Textbook Problems

A useful starting reference on introductory abstract algebra is *Artin's Algebra* (it's kind of dense though, so ask me for other recommendations if you want them). Several problems below were taken from it.

1. Characterize the abelian group $(\mathbb{F}_q, +)$ for each q .
2. Prove every field K of characteristic 0 admits a unique map $\mathbb{Q} \rightarrow K$.
3. How many automorphisms (self-isomorphisms) are there of $\mathbb{R}$?
4. ★ Some computations:
 - (a) Factor $x^9 - x$ and $x^{27} - x$ in $\mathbb{F}_3$.
 - (b) Factor $x^{16} - x$ over $\mathbb{F}_4$ and $\mathbb{F}_8$.
5. Given an irreducible polynomial $r(x)$ of degree k in $\mathbb{F}_p[x]$, does every root of $r(x)$ have to be a generator of $(\mathbb{F}_p[x]/(r(x)))^\times$?
6. ★ The *Frobenius* automorphism is the function $\phi : \mathbb{F}_q \rightarrow \mathbb{F}_q$ sending $x \rightarrow x^p$.
 - (a) Prove that ϕ is a field isomorphism.
 - (b) Consider any polynomial $p(x) \in \mathbb{F}_p[x]$. Prove that any automorphism of a (finite) field extension $\mathbb{F}_q$ of $\mathbb{F}_p$ must permute the roots of $p(x)$ in $\mathbb{F}_q$.
 - (c) Prove that there are at most $\log_p(q)$ automorphisms of $\mathbb{F}_q$, hence they are all generated by the Frobenius automorphism ϕ .
7. Which finite fields embed in other finite fields?
 - (a) Prove that for $q = p^k$, $r = p^l$, if $\mathbb{F}_q$ admits an injection into $\mathbb{F}_r$, then r is a power of q .
 - (b) Prove that $\mathbb{F}_r$ admits a unique subfield isomorphic to $\mathbb{F}_q$.
8. ★ As discussed earlier, every irreducible polynomial in $\mathbb{F}_p[x]$ of degree k divides $x^{p^k} - x$. We now investigate this further:
 - (a) Prove that if an irreducible polynomial of degree k divides $x^{p^l} - x$, then $k \mid l$.
 - (b) Prove that the irreducible factors of $x^{p^k} - x$ over $\mathbb{F}_p[x]$ are precisely the irreducible polynomials of degree dividing k .
 - (c) Give a general formula for the number of irreducible polynomials of degree k in $\mathbb{F}_p[x]$.
9. ★ Let $\phi_d(x)$ be the d^{th} cyclotomic polynomial. Prove that the irreducible factors of $\phi_d(x)$ over $\mathbb{F}_p$ are all of the same degree k , and compute k in terms of d .
10. ★ How do we check when a polynomial has multiple roots?
 - (a) Given a field K and a polynomial $p(x) \in K[x]$, define the *formal derivative* $p'(x)$ by mapping each monomial ax^k to kax^{k-1} . Prove this formal derivative operation satisfies the Leibniz rule, i.e. $(fg)' = f'g + fg'$.
 - (b) Consider a field K and a polynomial $p(x) \in K[x]$. Prove that $p(x)$ has multiple roots in K iff it has nontrivial gcd with its derivative $p'(x)$.
 - (c) Prove that any irreducible polynomial $p(x) \in K[x]$, where K either has characteristic 0 or is a finite field, has no multiple roots in $K[x]/(p(x))$.

For further reference, see “separable extension”.

3.7.2 Contest Problems

1. A polynomial $P(x)$ is a *base- n polynomial* if it is of the form $a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0$, where each a_i is an integer between 0 and $n-1$ inclusive and $a_d > 0$. Find the largest positive integer n such that for any real number c , there exists at most one base- n polynomial $P(x)$ for which $P(\sqrt{2} + \sqrt{3}) = c$.
2. Let $n > 3$ be a positive integer satisfying $2^n + 1 = 3p$, where p is a prime. Let $s_0 = \frac{2^{n-2}+1}{3}$ and $s_i = s_{i-1}^2 - 2$ for $i > 0$. Show that $p \mid 2s_{n-2} - 3$.
3. Let a be a positive integer and p a prime divisor of $a^3 - 3a + 1$, with $p \neq 3$. Prove that p is of the form $9k + 1$ or $9k - 1$, where k is integer.
4. Let $p > 2$ be a prime number. $\mathbb{F}_p[x]$ is defined as the set of polynomials in x with coefficients in $\mathbb{F}_p$ (the integers modulo p with usual addition and subtraction), so that two polynomials are equal if and only if the coefficients of x^k are equal in $\mathbb{F}_p$ for each nonnegative integer k . For example, $(x+2)(2x+3) = 2x^2 + 2x + 1$ in $\mathbb{F}_5[x]$ because the corresponding coefficients are equal modulo 5.
Let $f, g \in \mathbb{F}_p[x]$. The pair (f, g) is called compositional if

$$f(g(x)) \equiv x^{p^2} - x$$

in $\mathbb{F}_p[x]$. Find, with proof, the number of compositional pairs.

5. Let p be an odd prime. An integer x is called a quadratic non-residue if p does not divide $x - t^2$ for any integer t .
Denote by A the set of all integers a such that $1 \leq a < p$, and both a and $4 - a$ are quadratic non-residues. Calculate the remainder when the product of the elements of A is divided by p .
6. ★ In the 75th Annual Putnam Games, participants compete at mathematical games. Patniss and Keeta play a game in which they take turns choosing an element from the group of invertible $n \times n$ matrices with entries in the field $\mathbb{Z}/p\mathbb{Z}$ of integers modulo p , where n is a fixed positive integer and p is a fixed prime number. The rules of the game are:
 - (1) A player cannot choose an element that has been chosen by either player on any previous turn.
 - (2) A player can only choose an element that commutes with all previously chosen elements.
 - (3) A player who cannot choose an element on his/her turn loses the game.
 Patniss takes the first turn. Which player has a winning strategy?

3.7.3 Extra Contest Problems

Here are a few more problems that aren't necessarily super related to today's lecture, but give some good practice with finite field arithmetic.

1. Let $F_0, F_1, \dots$ be the sequence of Fibonacci numbers, with $F_0 = 0, F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. For $m > 2$, let R_m be the remainder when the product $\prod_{k=1}^{F_m-1} k^k$ is divided by F_m . Prove that R_m is also a Fibonacci number.
2. Positive integers $a_1, a_2, \dots, a_7, b_1, b_2, \dots, b_7$ satisfy $2 \leq a_i \leq 166$ and $a_i^{b_i} \cong a_{i+1}^2 \pmod{167}$ for each $1 \leq i \leq 7$ (where $a_8 = a_1$). Compute the minimum possible value of $b_1 b_2 \dots b_7 (b_1 + b_2 + \dots + b_7)$.
3. Let $\mathbb{N}$ denote the natural numbers. Compute the number of functions $f : \mathbb{N} \rightarrow \{0, 1, \dots, 16\}$ such that

$$f(x+17) = f(x) \quad \text{and} \quad f(x^2) \equiv f(x)^2 + 15 \pmod{17}$$

for all integers $x \geq 1$.

4. We define $\mathbb{F}_{101}[x]$ as the set of all polynomials in x with coefficients in $\mathbb{F}_{101}$ (the integers modulo 101 with usual addition and subtraction), so that two polynomials are equal if and only if the coefficients of x^k are equal in $\mathbb{F}_{101}$ for each nonnegative integer k . For example, $(x+3)(100x+5) = 100x^2 + 2x + 15$ in $\mathbb{F}_{101}[x]$ because the corresponding coefficients are equal modulo 101.

We say that $f(x) \in \mathbb{F}_{101}[x]$ is *lucky* if it has degree at most 1000 and there exist $g(x), h(x) \in \mathbb{F}_{101}[x]$ such that

$$f(x) = g(x)(x^{1001} - 1) + h(x)^{101} - h(x)$$

in $\mathbb{F}_{101}[x]$. Find the number of lucky polynomials.

5. Let $\mathbb{F}_p$ be the set of integers modulo p . Call a function $f : \mathbb{F}_p^2 \rightarrow \mathbb{F}_p$ quasiperiodic if there exist $a, b \in \mathbb{F}_p$, not both zero, so that $f(x+a, y+b) = f(x, y)$ for all $x, y \in \mathbb{F}_p$. Find the number of functions $\mathbb{F}_p^2 \rightarrow \mathbb{F}_p$ that can be written as the sum of some number of quasiperiodic functions.
6. Let $p = 2017$ be a prime and $\mathbb{F}_p$ be the integers modulo p . A function $f : \mathbb{Z} \rightarrow \mathbb{F}_p$ is called good if there is $\alpha \in \mathbb{F}_p$ with $\alpha \not\equiv 0 \pmod{p}$ such that

$$f(x)f(y) = f(x+y) + \alpha^y f(x-y) \pmod{p}$$

for all $x, y \in \mathbb{Z}$. How many good functions are there that are periodic with minimal period 2016?

7. The rows and columns of a $2^n \times 2^n$ table are numbered from 0 to $2^n - 1$. The cells of the table have been coloured with the following property being satisfied: for each $0 \leq i, j \leq 2^n - 1$, the j -th cell in the i -th row and the $(i+j)$ -th cell in the j -th row have the same colour. (The indices of the cells in a row are considered modulo 2^n .) Prove that the maximal possible number of colours is 2^n .
8. Let $k \in \mathbb{Z}^+$ and set $n = 2^k + 1$. Prove that n is a prime number if and only if the following holds: there is a permutation $a_1, \dots, a_{n-1}$ of the numbers $1, 2, \dots, n-1$ and a sequence of integers $g_1, \dots, g_{n-1}$, such that n divides $g_i^{a_i} - a_{i+1}$ for every $i \in \{1, 2, \dots, n-1\}$, where we set $a_n = a_1$.